Conditional Generation with Classically Trainable IQP Circuits

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Conditional generative models aim to represent a family of probability laws with one model, but a quantum implementation can instead hide the condition in a recompiled or reprogrammed circuit. We ask whether one compiled instantaneous quantum polynomial-time (IQP) circuit can achieve conditional control when the condition changes only its product-state input and its gate sequence and trained phases remain fixed. Our construction is classically trainable in a narrow sense: its selected Pauli observables, Hamming-kernel Conditional maximum mean discrepancy loss, and phase gradients are classically estimable, although we provide no efficient classical method for evaluating or sampling its full output law. Specializing established circuit-Fourier and IQP observable results shows how the fixed encoding limits the circuit's conditional response. We test this mechanism on Gaussian Hamming-weight targets by exactly evaluating 1,020 atomic models across four system sizes, three target widths, and ten paired seeds. Paired profile-total-variation-distance intervals support conditional control over an unconditioned model in 12 of 12 size–width cells. However, at unseen interior conditions, the shared circuit outperforms a target-aware adjacent-model lookup in 0 cells, gives mixed evidence in 2, and is unsupported in 10. Thus, product-state conditioning reliably controls this target family, but does not outperform a strong bank of condition-specific models; the results delineate the promise and cost of imposing one fixed deployed quantum circuit.

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1 Introduction

Conditional generative models represent a family $q_\theta(x \mid c)$ rather than a single probability law. In a quantum circuit Born machine, a condition can be introduced by recompiling the circuit, modulating variational gates, or changing the prepared input. These choices are operationally different. A recompiled model may use condition-specific programs, whereas a fixed deployment exposes only a data-loading interface. This paper studies the latter setting: every condition uses one ordered gate sequence and one trained phase vector, and only the initial product state changes.

This setting separates conditional control from classical access. A fixed quantum circuit may respond usefully to a classical condition even when its full bitstring law is not efficiently available classically. Conversely, selected expectation values and their gradients may be estimated probabilistically without implying efficient strong simulation or sampling of the output distribution [1, 2]. Sampling-hardness statements for IQP circuits require additional complexity assumptions and are not consequences of the experiments reported here [3–5].

Our construction encodes a scalar condition by broadcast R_Y rotations, then applies one complete two-local IQP circuit. The product input supplies a tractable reference for selected Pauli estimators, while the Hamming radial basis kernel admits a character-mask representation. Established quantum data-encoding results imply finite Fourier response support [6–8]; the active-generator expansion of IQP observables further filters which frequencies enter a selected training signal [9]. We present these facts as explanatory specializations and syntheses, not as novel central theorems.

The paper makes three scoped contributions.

First, it gives a precise fixed-circuit definition of product-state-conditioned IQP generation and states exactly what remains invariant at deployment. Second, it delineates the accessible classical training signals from the inaccessible full output law. Third, it provides a broad Gaussian Hamming-profile characterization: $n \in \{8, 12, 16, 19\}$, three widths, ten paired seeds, explicit no-condition and independent-peak baselines, eight common held-out coordinates, nested condition grids, two data-allocation regimes, and compact topology, encoding, and ancilla ablations. Every cell and interval is reported; there is no empirical success gate.

We use *conditional generalization* once in a deliberately narrow sense: interpolation to conditions unseen during training but strictly inside the training interval. We use *interpolation* hereafter. Extrapolation, broader distribution shift, universal conditional expressivity, and quantum advantage are outside scope.

2 Related work and scope of classical trainability

Diagonal commuting circuits occupy a useful boundary between restricted classical simulation and sampling-hardness results [2, 5, 10]. Probabilistic simulation can estimate particular observables of such circuits under appropriate input and observable structure [1]. Recent work develops this idea into scalable classical optimization tools for IQP circuit Born machines [9, 11]. Our use is narrower than a generic trainability claim: it refers to the declared observables, estimator budgets, objective, and gradients in Sec. 4.

Parameterized quantum circuits with data reuploading generate Fourier series whose frequency content is controlled by the encoding Hamiltonians [6–8]. We use this established response language to explain what condition dependence a fixed product-input circuit can and cannot represent. Kernel two-sample objectives provide a distributional training signal without requiring pointwise likelihoods [12]. The contribution here is the fixed-circuit conditional formulation and its systematic Gaussian performance map, not a new kernel or Fourier theorem.

Quantum circuit Born machines have previously been trained by kernel MMD, and conditional Born machines have encoded external variables for event-generation tasks [13, 14]. Prior QCBM generalization studies ask whether a learned law produces unseen valid or high-quality bitstrings [15]. Our interpolation question is different and narrower: the output law is evaluated at unseen values of an explicit Condition coordinate while the compiled circuit and phase vector stay fixed. We do not infer broader sample- level or distribution-shift generalization from that test.

The distinction matters because favorable finite-budget estimators do not guarantee a favorable optimization landscape [16, 17]. Nor does a low discrepancy in Hamming weight imply a low discrepancy over all bitstrings. Our benchmark is constructed to make both limitations visible.

3 Product-state-conditioned IQP model

Let $g_j \in \{0,1\}^n$ identify a diagonal Pauli string $Z_{g_j} = \bigotimes_i Z_i^{g_{j,i}}$. For a fixed ordered generator bank $G = (g_1, \ldots, g_m)$ and phases $\theta \in \mathbb{R}^m$, define

$$D_Z(\theta) = \prod_{j=1}^{m} \exp(i\theta_j Z_{g_j}). \tag{1}$$

For a condition c, the output law is

$$q_\theta(x \mid c) = \left| \langle x|H^{\otimes n} D_Z(\theta) H^{\otimes n} \times \bigotimes_{i=1}^{n} R_Y(\alpha_i(c))|0\rangle \right|^2. \tag{2}$$

A model is *fixed-circuit conditional* when the compiled sequence, ordered G, and θ are identical for every condition; only the initial angles may vary. In the main campaign,

$$\alpha_i(c) = \pi(0.35 - 0.70c), \qquad i = 1, \ldots, n. \tag{3}$$

The ansatz contains all one- and two-qubit diagonal generators, hence $m = n(n+1)/2$ phases. The implementation hashes the generator matrix and checkpoint and verifies their invariance across all evaluated conditions.

Two controls clarify what is gained by conditioning. The parameter-matched *no-condition* model receives the neutral input $c = 1/2$ for every target. The *single-peak bank* trains one independent circuit at each training condition. At a held-out coordinate, we compare against (i) target-aware selection of the adjacent bank output with

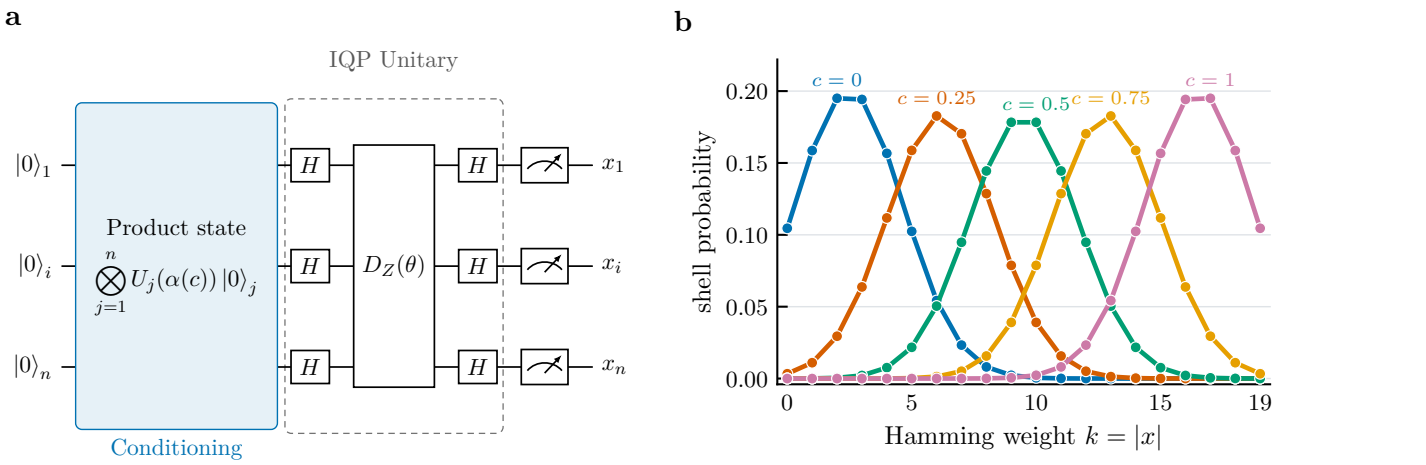


Figure 1: Product-state conditioning and the Gaussian target family. (a) Conditioning prepares the product input $\bigotimes_i U_i(\alpha(c))\,|0\rangle_i$; the IQP unitary and phase vector are shared across conditions. (b) Five representative 19-qubit target profiles; probability is uniform within each Hamming shell.

lower profile TVD and (ii) a target-blind barycentric mixture of those outputs. If $c_L < c < c_R$, the mixture is

$$q_{\text{mix}}(x \mid c) = \frac{c_R - c}{c_R - c_L} q_L(x) + \frac{c - c_L}{c_R - c_L} q_R(x). \quad (4)$$

Its weights use coordinates only; target metrics never enter. The adjacent lookup is intentionally generous because it may inspect the held-out target to choose between q_L and q_R.

4 Classical training signals

For the Hamming radial basis kernel $k_\sigma(x, y) = \exp[-d_H(x, y)/(2\sigma^2)]$, draw a mask $a \in \{0, 1\}^n$ with independent coordinates

$$a_i \sim \text{Bernoulli}(q_\sigma), \qquad q_\sigma = \frac{1 - e^{-1/(2\sigma^2)}}{2}. \quad (5)$$

Then

$$k_\sigma(x, y) = \mathbb{E}_a[(-1)^{a\cdot x}(-1)^{a\cdot y}]. \quad (6)$$

Thus squared MMD is an average of Pauli-character discrepancies. Independent model and reference draws yield the standard unbiased U-statistic [12]; differentiation of its finite sampled form gives phase gradients. We average bandwidths $(1, 2, 4)$ and conditions. Operator masks, model/reference samples, target samples, mini-batches, and validation streams have separately derived seeds.

This establishes access to the declared training signal, not to every output probability. The production optimizer uses 64 masks and 8,192 model/reference samples per bandwidth in training, and 256 masks and 16,384 samples in validation. Exact probabilities are enumerated only after checkpoint selection for evaluation. At $n = 19$ this controlled audit is feasible, but it is not an efficient asymptotic full-distribution procedure.

5 Established response structure

The following propositions collect established structure in the notation of Eq. (2); derivations appear in the supplement.

Proposition 1 (Affine product-input support)**.** *For fixed G, θ, b, W and output x, if $\alpha(c) = b + Wc$, the support of $c \mapsto q_\theta(x \mid c)$ is contained in*

$$\Omega_{\text{full}}(W) = \left\{ \sum_{i=1}^{n} s_i w_i : s_i \in \{-1, 0, 1\} \right\}, \quad (7)$$

where w_i^T is row i of W.

This is a direct specialization of multivariate circuit-Fourier theory [6–8]. The inclusion can be strict because frequencies coincide or coefficients cancel, and it gives no efficient procedure for evaluating the coefficients.

Proposition 2 (Observable filter)**.** *For $F_a(c) = \langle Z_a \rangle_c$, only generators satisfying $a \cdot g_j = 1 \pmod 2$ enter the IQP active-generator expansion. Group their subsets by XOR mask h. A term survives on the real R_Y product input only if $h \wedge a = 0$, and its condition frequencies are signed sums of rows w_i with $i \in \text{supp}(a) \cup \text{supp}(h)$.*

This combines the established active-generator formula [9] with $\langle Y_i \rangle = 0$ on a real product state. Selected observable responses can therefore occupy a smaller space than full probabilities. At fixed encoding width both spaces are finite dimensional, which precludes a claim of universal density in arbitrary continuous conditional laws.

6 Gaussian benchmark and frozen protocol

The target is exchangeable within each Hamming shell. At system size n and condition $c \in [0, 1]$, the requested mean is

$$\mu_n(c) = n(0.15 + 0.70c), \tag{8}$$

and the nominal width is $s\sqrt{n}$ for $s \in \{0.35, 0.50, 0.75\}$. We discretize a Gaussian over weights $0, \ldots, n$ and recalibrate its untruncated location numerically so the truncated distribution has mean $\mu_n(c)$. Its shell probability is then distributed uniformly over the $\binom{n}{k}$ strings at weight k. Profile TVD, the TVD between shell-mass vectors, is the primary metric; raw bitstring TVD, mean absolute error, width absolute error, and exact Conditional MMD are secondary diagnostics.

The factorial crosses $n \in \{8, 12, 16, 19\}$, three widths, and ten matched optimizer/data seeds. Shared conditioned and no-condition models train on $c \in \{0, 1/4, 1/2, 3/4, 1\}$; five single-peak circuits form the independent bank. All shared models are evaluated at those five coordinates and the common unseen grid $c \in \{1/16, 3/16, \ldots, 15/16\}$. No held-out samples are generated before exact post-selection evaluation.

Every model receives 2,000 Adam updates with cosine learning-rate decay $0.01 \to 0.001$. The total target minibatch contains 450 strings, balanced across conditions; an independent model also uses 450 samples per update. Validation occurs every 25 updates, and checkpoint selection uses only minimum validation Conditional MMD. Shared and no-condition models have matched initialization and estimator streams. Bank streams are tagged by peak. The factorial contributes 840 atomic identities.

The condition-count study fixes $n = 19$ and $s = 0.50$. Nested grids with $K \in \{2, 3, 5, 9\}$ are evaluated under 10,000/2,000 training/validation draws per peak and under fixed totals of 18,000/3,600 divided among peaks. Each target has one deterministic 10,000/2,000 master pool; fixed-total datasets are prefixes. A nine-peak bank supplies a deliberately generous capacity reference, with nested checkpoints reused only by declared identity. Together with the factorial, the frozen manifest declares exactly 1,020 atomic models.

A 100-update $n = 19, K = 9$ resource preflight recorded 22.02 seconds and 1.96 GB peak resident memory on the execution host, projecting 440.30 seconds per model. This passed the preregistered one-hour and 28 GiB resource limits. These are resource checks, not performance thresholds. Full manifest, dataset, stream, checkpoint, probability, environment, runtime, source, and revision digests are archived.

7 Conditional control and interpolation

We average conditions within each optimizer seed before method comparison and bootstrap the ten paired seed effects 10,000 times with seed 20260811. A strictly positive 95% interval is described as supported, a strictly negative interval as unsupported, and an interval containing zero as mixed. These are descriptive labels, not a publication gate. Every one of the 12 cells remains visible.

For the representative $n = 19, s = 0.50$ cell, no-condition minus conditioned training profile TVD is 0.2668 [0.2587, 0.2737]; held-out lookup minus conditioned profile TVD is 0.0016 [-0.0012, 0.0043]; and held-out barycentric mixture minus conditioned profile TVD is -0.0029 [-0.0053, -0.0006]. The conditioned-minus-single-peak training fidelity gap is 0.0878 [0.0860, 0.0897] (positive values mean a cost of sharing). The corresponding raw-bitstring lookup effect is -0.0606 [-0.0749, -0.0461]. Across the full surface, conditional-control profile intervals are supported, mixed, and unsupported in 12, 0, and 0 cells, respectively. Interpolation against lookup has counts 0/2/ 10, while interpolation against the target-blind mixture has counts 2/1/ 9.

8 Performance across size, width, and peak location

Figure 3 resolves the aggregate statements into the complete 4×3 performance surface. Positive

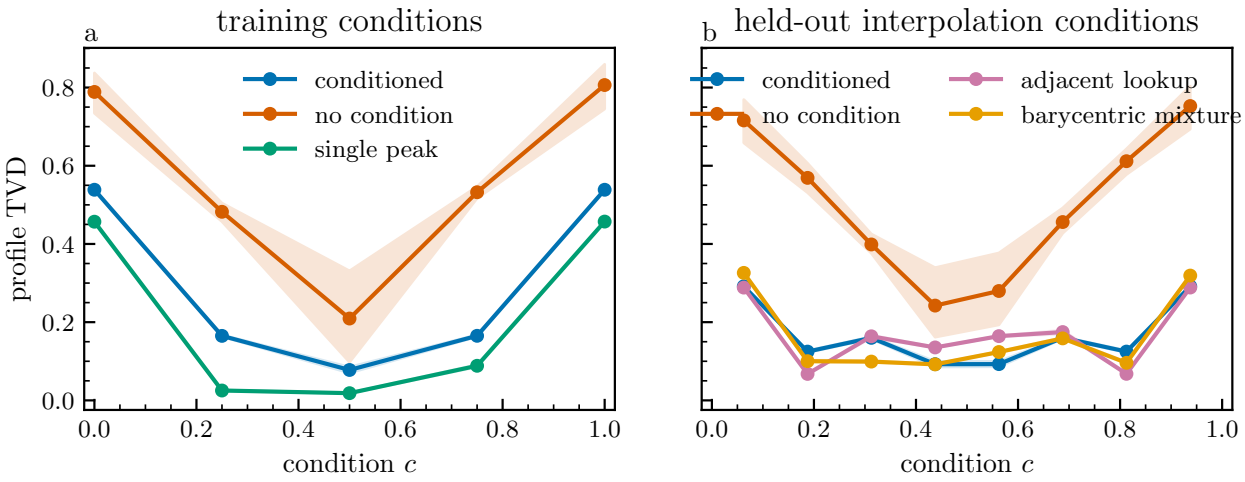


Figure 2: Training-condition control and held-out interpolation at $n = 19, s = 0.50$. Lines are ten-seed means and bands are deterministic 95% seed-bootstrap intervals. The held-out panel compares the conditioned circuit with no conditioning, target-aware adjacent lookup, and the coordinate-only barycentric mixture.

Table 1: Claim summary over the 12 size–width cells. Counts denote supported / mixed / unsupported paired 95% intervals; positive effects favor the conditioned model. No count is used as a success threshold.

Comparison	Supported	Mixed	Unsupported
Training profile TVD: no condition − conditioned	12	0	0
Held-out profile TVD: lookup − conditioned	0	2	10
Held-out profile TVD: mixture − conditioned	2	1	9
Training bitstring TVD: no condition − conditioned	12	0	0
Held-out bitstring TVD: lookup − conditioned	2	1	9

and negative cells use the same color scale, and the numerical paired effect is printed in each cell. The per-condition curves and exhaustive intervals in the supplement reveal whether an aggregate seed mean masks a difficult peak location. In particular, the endpoints probe truncation asymmetry, while the common odd-sixteenth grid makes held-out comparisons identical across all training grids.

The width sweep changes two distinct difficulties. Narrow profiles demand accurate concentration in weight, while wider profiles allocate mass to more shells. Increasing n also increases both the output space and the complete two-local parameter count. We therefore avoid pooling cells across n or s into one headline effect. The heatmaps summarize direction and magnitude; the table in the supplement retains the paired interval and cell-level wins.

9 Condition-count and data-allocation scaling

The K study asks whether adding training conditions helps because the model sees the conditional curve more densely or merely because it receives more data. The fixed-per-peak regime increases total samples with K; the fixed-total regime holds total samples constant. Because all subsets are prefixes of the same target-specific master pools, differences cannot be attributed to unrelated draws. We report the per-condition and worst- condition profile TVD rather than only the overall mean.

The shared circuit always has 190 phases and 2,000 optimizer updates. An independent bank has $190K$ stored phases and $2,000K$ optimizer updates. Its larger optimization work is part of the baseline definition, not hidden from the comparison. The nine-peak bank is particularly generous: its nested outputs are reused for all K values, whereas shared circuits are retrained for

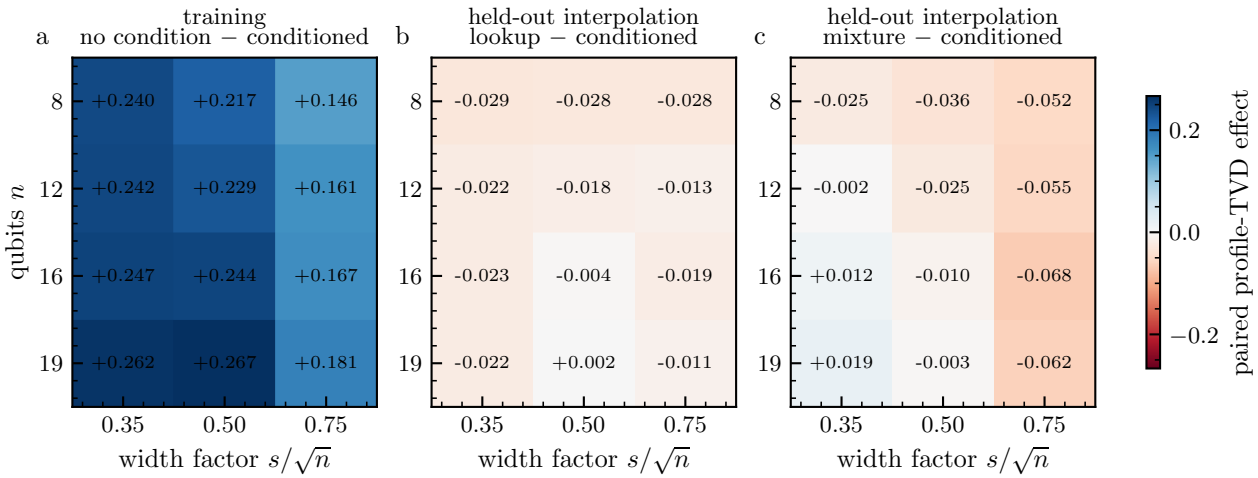


Figure 3: Complete Gaussian performance surface. Each entry is a ten-seed paired profile-TVD effect: no-condition minus conditioned at training conditions, and adjacent-lookup or barycentric- mixture minus conditioned at held-out conditions. Positive values favor the conditioned circuit.

their declared grid and data regime.

10 Compact Gaussian ablations

Three archived studies test modeling choices outside the frozen factorial. First, complete two-local connectivity is compared with lattice and heavy-hex generator banks on a two-peak 19-qubit task. Second, the broadcast-angle map is compared with a symmetric-reupload encoding at matched parameter count. Third, ancilla-assisted models are compared with visible-only circuits whose additional higher-order generators match the parameter counts. Figure 5 reports these studies descriptively and separately from the 1,020-model campaign because their target grids and training protocols are not identical.

The ancilla comparison is especially easy to overinterpret. Adding ancillas changes the effective visible marginal family, whereas adding visible-only generators changes capacity without changing the measured subsystem. A difference between them is evidence about this benchmark and construction, not a general ancilla advantage. Target-symmetry-aware post-processing is analyzed separately in the supplement and is excluded from every raw-circuit claim.

11 Limitations, cost, and reproducibility

The target family is one-dimensional and exchangeable. Low profile TVD means the model places the right total mass in each Hamming shell; it does not mean that probability is uniform among strings inside a shell. Raw bitstring TVD is therefore reported in every cell and is the main representational limitation. The benchmark demonstrates interpolation inside a fixed interval, not extrapolation, robustness to shifted target families, or universal conditional generation.

Our operational classical-trainability statement is likewise bounded. The training objective and gradients use finite classical estimators, but exact post-selection evaluation enumerates 2^n probabilities. Runtime grows with system size, connectivity, estimator budget, and optimizer work. Nothing here establishes efficient classical sampling of the trained output, a quantum advantage, or a favorable asymptotic optimization landscape.

The campaign is manifest-driven and immutable. It declares exactly 1,020 run identities, rejects undeclared or duplicate completion, binds datasets and estimator streams by role, and refuses partial bank reuse. Execution is resumable and shardable. Each result binds the manifest, source tree, environment, revision, dataset, checkpoint, probability archive, invocation, run-

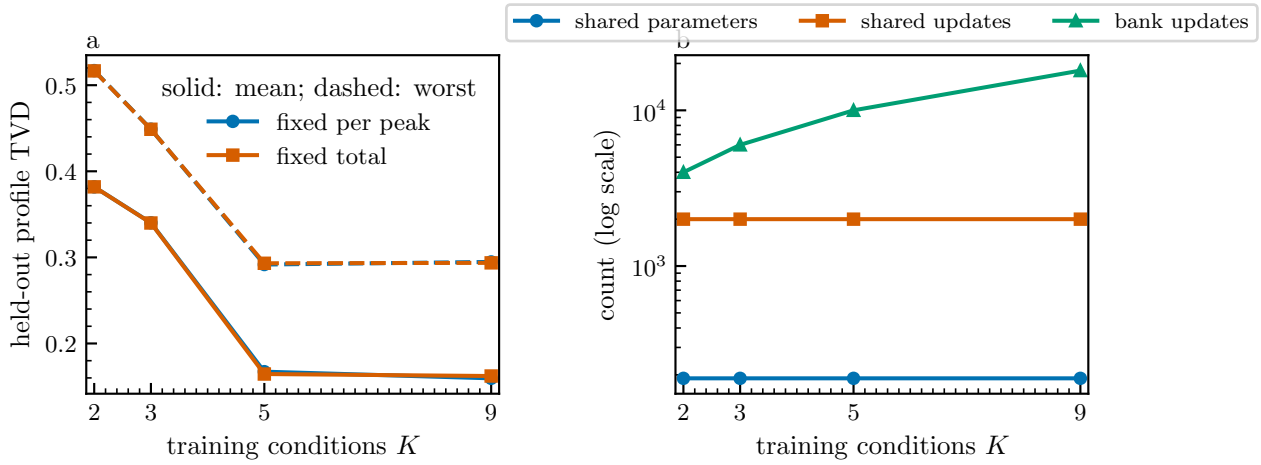


Figure 4: Condition-count characterization at $n = 19, s = 0.50$. (a) Mean and worst-condition held-out profile TVD for the fixed-per-peak and fixed-total regimes. (b) Shared-model parameter count and optimization work compared with the independent bank.

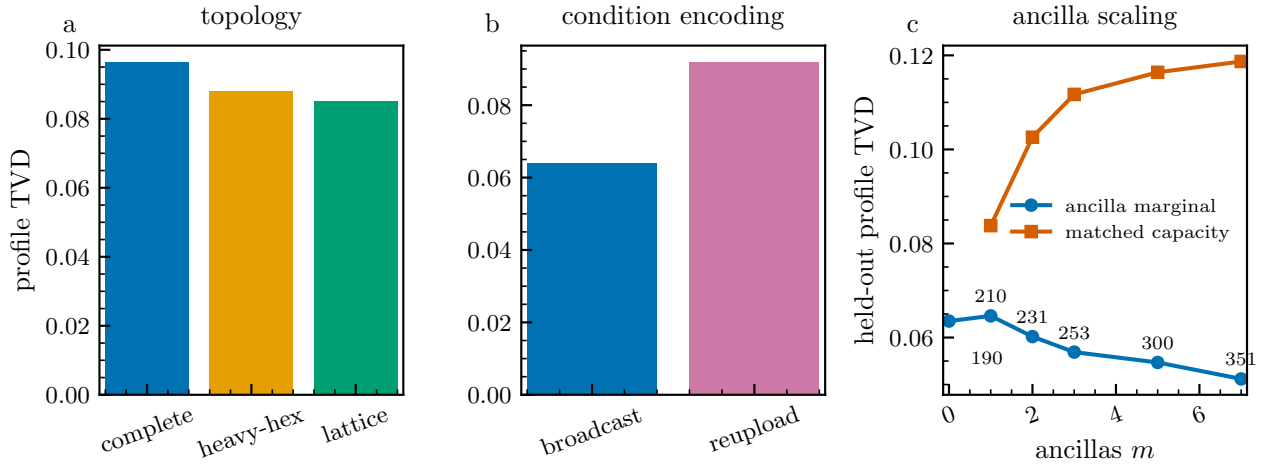


Figure 5: Compact archived Gaussian ablations. (a) Complete, lattice, and heavy-hex topologies. (b) Broadcast-angle and symmetric- reupload encodings. (c) Ancilla scaling; annotations give parameter counts. Protocols differ from the frozen factorial, so panels are descriptive rather than pooled campaign evidence.

time, and peak memory. Figures, tables, and claims are generated only from the complete report. The supplement describes regeneration, archive checks, and clean-build verification.

12 Conclusion

Product-state conditioning gives a precise fixed-deployment route to conditional IQP generation: one compiled commuting circuit and one phase vector respond to a classical condition through local input rotations. The selected Hamming-kernel training objective and gradients remain accessible to classical probabilistic estimation, while full-output access is not claimed. Across the Gaussian campaign, conditioning improves training-condition profile TVD in all 12 size–width cells. Held-out outputs, however, usually do not outperform the generous adjacent single-peak lookup or its target-blind barycentric mixture: the lookup comparison is mixed in two cells and unsupported in ten, while the mixture comparison is supported in two, mixed in one, and unsupported in nine. The result is therefore evidence for robust conditional control, not a broad claim of superior interpolation. This positive-and-negative performance map provides a transparent foundation for improving fixed-circuit conditional models while keeping the present claim restricted to the tested Gaussian Hamming-profile family.

Data and code availability

The source repository contains the frozen manifest, role-tagged datasets, immutable ledgers, checkpoints, exact probability archives, report, claim registry, figure/table generators, and clean-build scripts. The canonical PDF is promoted only after all declared identities and evaluations complete.

Release metadata

Author names, affiliations, individual contributions, competing interests, arXiv category and license, and final human proof approval remain to be supplied and verified by the research team. This document is not upload-ready. No upload or submission is automated.

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A Analytical details

This appendix derives the response and estimator statements used in the main text. They specialize established circuit-Fourier and probabilistic-IQP results to product-state-conditioned notation and are not presented as novel central theorems.

A.1 Full-probability response support

One encoded qubit has density operator

$$\begin{aligned}\rho_i(\alpha_i) &= R_Y(\alpha_i)|0\rangle\langle 0|R_Y(\alpha_i)^\dagger \\ &= \tfrac{1}{2}\left(I + \sin\alpha_i X + \cos\alpha_i Z\right).\end{aligned} \tag{9}$$

Each entry is a linear combination of angle frequencies $-1, 0, +1$. Tensoring and substituting $\alpha_i(c) = b_i + w_i^\mathsf{T} c$ gives fixed matrices A_s such that

$$\rho(c) = \sum_{s\in\{-1,0,1\}^n} A_s e^{i\sum_i s_i b_i} e^{i(\sum_i s_i w_i)^\mathsf{T} c}. \tag{10}$$

For fixed unitary U and outcome x, $q(x \mid c) = \mathrm{Tr}[U^\dagger|x\rangle\langle x|U\rho(c)]$ is a fixed linear functional of $\rho(c)$ and introduces no new condition frequency. This proves Eq. (7). Zero rows, coincident sums, and coefficient cancellations can only shrink the support. The trace coefficient can still require exponential work to evaluate.

For the broadcast scalar encoding in Eq. (3), every row of W is identical. Frequencies therefore collapse to integer multiples of the single slope, with multiplicities generated by different sign patterns. This degeneracy is useful for a one-coordinate location target but also limits the independent conditional directions available at fixed width.

A.2 Pauli observable filter

The final Hadamards map measured Z_a to X_a. A diagonal generator Z_{g_j} satisfying $a\cdot g_j = 0 \pmod 2$ commutes with X_a and cancels. For an active generator,

$$e^{-i\theta_j Z_{g_j}} X_a e^{i\theta_j Z_{g_j}} = X_a e^{2i\theta_j Z_{g_j}}. \tag{11}$$

Since the diagonal generators commute,

$$\begin{aligned}D_Z(\theta)^\dagger X_a D_Z(\theta) = X_a \prod_{j:a\cdot g_j=1} &[\cos(2\theta_j) I \\ &+ i\sin(2\theta_j) Z_{g_j}].\end{aligned} \tag{12}$$

Expanding groups active subsets by $h = \bigoplus_{j\in S} g_j$. Conjugating $X_a Z_h$ through the first Hadamards produces local $Z_a X_h$ factors. On the real product input,

$$\langle X_i\rangle = \sin\alpha_i, \qquad \langle Z_i\rangle = \cos\alpha_i, \qquad \langle Y_i\rangle = 0. \tag{13}$$

Every term with $h\wedge a \neq 0$ therefore vanishes. Surviving terms factor into sines and cosines on $\mathrm{supp}(a)\cup\mathrm{supp}(h)$; expanding into exponentials yields the signed sums stated in the proposition.

A.3 Hamming-kernel mask identity and estimator

For one bit, let $a \sim \mathrm{Bernoulli}(q)$. Then

$$\mathbb{E}_a[(-1)^{a(x-y)}] = \begin{cases} 1, & x = y, \\ 1-2q, & x \neq y. \end{cases} \tag{14}$$

Choosing $q = q_\sigma$ from Eq. (5) gives $1-2q = e^{-1/(2\sigma^2)}$. Independence across bits yields Eq. (6). Let $\phi_a(x) = (-1)^{a\cdot x}$. For target samples $y_1,\ldots,y_T$ and model samples $x_1,\ldots,x_R$, the fixed-mask unbiased squared

Table 2: Frozen data allocation in the condition-count study.

K	fixed per peak		fixed total	
	train	validation	train	validation
2	10,000	2,000	9,000	1,800
3	10,000	2,000	6,000	1,200
5	10,000	2,000	3,600	720
9	10,000	2,000	2,000	400

discrepancy is

$$\begin{aligned}\widehat{\Delta}_a^2 = &\frac{1}{R(R-1)}\sum_{r\neq r'}\phi_a(x_r)\phi_a(x_{r'}) \\ &+\frac{1}{T(T-1)}\sum_{t\neq t'}\phi_a(y_t)\phi_a(y_{t'}) \\ &-\frac{2}{RT}\sum_{r,t}\phi_a(x_r)\phi_a(y_t).\end{aligned} \tag{15}$$

Averaging over masks, bandwidths, and conditions gives sampled Conditional MMD. Differentiation acts only on the model expectation because target and mask laws do not depend on phase parameters. This gives estimator access, not a lower bound on gradient magnitude or a convergence guarantee.

After checkpoint selection, the exact population value is computed from Walsh–Hadamard coefficients. With $\widehat{p}(a) = \sum_x p(x)(-1)^{a\cdot x}$,

$$\mathrm{MMD}_\sigma^2(p,q) = \sum_{a\in\{0,1\}^n} q_\sigma^{|a|}(1-q_\sigma)^{n-|a|}\left[\widehat{p}(a)-\widehat{q}(a)\right]^2. \tag{16}$$

B Target construction and data isolation

For a nominal location ℓ and width $s\sqrt{n}$, define unnormalized shell mass

$$r_k(\ell,s) = \exp\left[-\frac{(k-\ell)^2}{2s^2n}\right], \qquad k=0,\dots,n. \tag{17}$$

Truncation changes the realized mean, so a monotone scalar root solve chooses ℓ such that $\sum_k k r_k / \sum_k r_k = \mu_n(c)$. The resulting normalized shell mass $P(k \mid c)$ induces

$$p(x \mid c) = \frac{P(|x| \mid c)}{\binom{n}{|x|}}. \tag{18}$$

This is an exact analytic law. Training nonetheless sees only finite sampled strings.

Each combination of n, width factor, and master seed has one immutable data archive. It stores the nine-condition master grid, role-tagged train and validation strings, derived random seeds, and hashes. The main five-condition grid selects even master indices. The $K = 2, 3, 5$ grids are nested subsets of the nine-grid conditions. Fixed-total arrays are exact prefixes of the same 10,000/2,000 per-target masters. The archive schema has no held-out role, and the loader rejects any such role before training.

C Complete factorial results

Table 3 reports every size–width cell. Effects average conditions within seed and then pair by seed. Positive values favor the conditioned model. Status letters are S (95% interval strictly positive), M (contains zero), and U (strictly negative), ordered as training control, held-out adjacent lookup, and held-out barycentric mixture. They describe the observed uncertainty and do not implement a success rule.

For auditability, the machine-readable report also records cell-level wins at each seed–condition pair, mean and width error, exact Conditional MMD, checkpoint selection, and model and target shell probabilities. The complete probability artifact retains all 2^n model probabilities at every declared evaluation coordinate.

Table 3: All factorial profile-TVD effects with deterministic paired 95% bootstrap intervals.

n	$s/\sqrt{n}$	no condition − conditioned	lookup − conditioned	mixture − conditioned	S/M/U
8	0.35	+0.2402 [+0.2278, +0.2536]	-0.0292 [-0.0468, -0.0108]	-0.0248 [-0.0418, -0.0064]	S/U/U
8	0.50	+0.2170 [+0.2015, +0.2418]	-0.0280 [-0.0336, -0.0225]	-0.0363 [-0.0432, -0.0292]	S/U/U
8	0.75	+0.1460 [+0.1293, +0.1714]	-0.0276 [-0.0339, -0.0214]	-0.0525 [-0.0584, -0.0461]	S/U/U
12	0.35	+0.2419 [+0.2352, +0.2505]	-0.0221 [-0.0255, -0.0190]	-0.0016 [-0.0052, +0.0019]	S/U/M
12	0.50	+0.2291 [+0.2213, +0.2391]	-0.0180 [-0.0213, -0.0146]	-0.0251 [-0.0283, -0.0216]	S/U/U
12	0.75	+0.1608 [+0.1515, +0.1722]	-0.0129 [-0.0151, -0.0107]	-0.0548 [-0.0567, -0.0526]	S/U/U
16	0.35	+0.2472 [+0.2421, +0.2519]	-0.0229 [-0.0279, -0.0174]	+0.0117 [+0.0067, +0.0172]	S/U/S
16	0.50	+0.2440 [+0.2354, +0.2537]	-0.0040 [-0.0071, +0.0001]	-0.0099 [-0.0133, -0.0057]	S/M/U
16	0.75	+0.1667 [+0.1570, +0.1770]	-0.0192 [-0.0205, -0.0178]	-0.0681 [-0.0696, -0.0666]	S/U/U
19	0.35	+0.2620 [+0.2594, +0.2645]	-0.0217 [-0.0240, -0.0192]	+0.0188 [+0.0164, +0.0213]	S/U/S
19	0.50	+0.2668 [+0.2587, +0.2737]	+0.0016 [-0.0012, +0.0043]	-0.0029 [-0.0053, -0.0006]	S/M/U
19	0.75	+0.1807 [+0.1648, +0.1999]	-0.0111 [-0.0165, -0.0049]	-0.0619 [-0.0670, -0.0560]	S/U/U

D Raw-bitstring diagnostics

Table 5 keeps the raw-law limitation adjacent to the primary profile result. Because targets are uniform within shells, profile TVD is a quotient metric: it ignores redistribution within a shell. The difference between profile and bitstring TVD quantifies that unmodeled structure. A positive paired effect still means the conditioned model is closer than the named baseline, but a large absolute TVD remains a limitation even when the relative comparison is favorable.

E Condition-count results

Table 7 reports both data regimes, the average and worst held-out coordinate, parameter count, and optimization work. The shared-model parameter and update counts do not grow with K; the independent bank's stored parameters and optimizer work grow linearly. The per-condition JSON rows behind this table are retained so that a mean cannot hide a single difficult coordinate.

Table 9 exposes every held-out coordinate rather than leaving those values only in the machine-readable report. Entries are ten-seed means of profile TVD; lower is better.

Table 9: Per-condition held-out profile TVD in the condition-count study.

K	Regime	c	Conditioned	No condition	Lookup	Mixture
2	fixed per peak	0.0625	0.1687	0.5621	0.2885	0.3119
2	fixed per peak	0.1875	0.4635	0.6671	0.0677	0.1739
2	fixed per peak	0.3125	0.5170	0.7480	0.3494	0.4125
2	fixed per peak	0.4375	0.3785	0.7775	0.5992	0.5613
2	fixed per peak	0.5625	0.3784	0.7790	0.5991	0.5613
2	fixed per peak	0.6875	0.5169	0.7499	0.3492	0.4125
2	fixed per peak	0.8125	0.4634	0.6855	0.0679	0.1741
2	fixed per peak	0.9375	0.1687	0.5752	0.2886	0.3121
3	fixed per peak	0.0625	0.1816	0.5632	0.2885	0.3341

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Table 4: Conditioned-model profile-TVD wins at the seed–Condition level in every factorial cell. Denominators are 50 Training comparisons and 80 Held-out comparisons per cell.

n	$s/\sqrt{n}$	vs. no condition	vs. lookup	vs. mixture
8	0.35	50/50	23/80	28/80
8	0.50	50/50	8/80	10/80
8	0.75	50/50	19/80	6/80
12	0.35	48/50	30/80	39/80
12	0.50	48/50	18/80	21/80
12	0.75	50/50	20/80	20/80
16	0.35	46/50	11/80	50/80
16	0.50	48/50	38/80	30/80
16	0.75	50/50	20/80	20/80
19	0.35	46/50	27/80	58/80
19	0.50	47/50	40/80	38/80
19	0.75	49/50	28/80	20/80

Table 5: Raw bitstring TVD across all factorial cells. The middle columns are the conditioned model's absolute mean TVD; effects are paired 95% intervals.

n	$s/\sqrt{n}$	training	held out	no condition − conditioned	lookup − conditioned
8	0.35	0.3394	0.2660	+0.2388 [+0.2232, +0.2554]	-0.0243 [-0.0445, -0.0054]
8	0.50	0.3086	0.2371	+0.2055 [+0.1855, +0.2306]	-0.0380 [-0.0491, -0.0277]
8	0.75	0.3229	0.2586	+0.1288 [+0.1018, +0.1699]	-0.0099 [-0.0165, -0.0032]
12	0.35	0.4179	0.3274	+0.2578 [+0.2494, +0.2684]	-0.0371 [-0.0466, -0.0271]
12	0.50	0.3661	0.2825	+0.2594 [+0.2388, +0.2782]	-0.0356 [-0.0470, -0.0249]
12	0.75	0.3596	0.2954	+0.1687 [+0.1409, +0.1999]	+0.0156 [+0.0083, +0.0233]
16	0.35	0.4618	0.3587	+0.2515 [+0.2406, +0.2632]	-0.0429 [-0.0538, -0.0319]
16	0.50	0.4061	0.3135	+0.2575 [+0.2402, +0.2765]	-0.0418 [-0.0510, -0.0318]
16	0.75	0.3929	0.3279	+0.2100 [+0.1710, +0.2486]	+0.0250 [+0.0190, +0.0313]
19	0.35	0.4964	0.3905	+0.2394 [+0.2256, +0.2523]	-0.0610 [-0.0776, -0.0458]
19	0.50	0.4380	0.3444	+0.2559 [+0.2370, +0.2723]	-0.0606 [-0.0749, -0.0461]
19	0.75	0.4292	0.3573	+0.1940 [+0.1476, +0.2405]	+0.0074 [-0.0032, +0.0165]

Table 6: Absolute secondary diagnostics for the conditioned model, averaged over Conditions and seeds. "Train" and "held" denote Training and Held-out coordinates.

n	$s/\sqrt{n}$	mean absolute error train	mean absolute error held	width absolute error train	width absolute error held	exact Conditional MMD train	exact Conditional MMD held
8	0.35	0.5797	0.2795	0.3714	0.3621	0.0184	0.0046
8	0.50	0.5964	0.3027	0.1428	0.1308	0.0193	0.0046
8	0.75	0.6319	0.3554	0.1332	0.2003	0.0208	0.0055
12	0.35	0.8565	0.3984	0.5225	0.5346	0.0204	0.0045
12	0.50	0.8969	0.4578	0.1463	0.1804	0.0220	0.0048
12	0.75	0.9641	0.5583	0.2829	0.3762	0.0244	0.0061
16	0.35	1.1360	0.5214	0.6114	0.6335	0.0206	0.0041
16	0.50	1.1909	0.6016	0.1426	0.2335	0.0228	0.0047
16	0.75	1.2908	0.7502	0.4486	0.5670	0.0259	0.0063
19	0.35	1.3490	0.6177	0.6631	0.6899	0.0204	0.0040
19	0.50	1.4049	0.7026	0.1503	0.2701	0.0228	0.0045
19	0.75	1.5071	0.8503	0.5448	0.6708	0.0263	0.0064

Table 7: Condition-count and data-allocation results at $n = 19, s = 0.50$.

K	regime	train/peak	val./peak	mean	worst	params	shared upd.	bank upd.
2	fixed per peak	10,000	2,000	0.3819	0.5170	190	2,000	4,000
3	fixed per peak	10,000	2,000	0.3402	0.4489	190	2,000	6,000
5	fixed per peak	10,000	2,000	0.1673	0.2917	190	2,000	10,000
9	fixed per peak	10,000	2,000	0.1595	0.2947	190	2,000	18,000
2	fixed total	9,000	1,800	0.3819	0.5166	190	2,000	4,000
3	fixed total	6,000	1,200	0.3399	0.4488	190	2,000	6,000
5	fixed total	3,600	720	0.1645	0.2932	190	2,000	10,000
9	fixed total	2,000	400	0.1622	0.2936	190	2,000	18,000

K	Regime	c	Conditioned	No condition	Lookup	Mixture
3	fixed per peak	0.1875	0.4140	0.5397	0.0677	0.2213
3	fixed per peak	0.3125	0.4473	0.5564	0.3494	0.2114
3	fixed per peak	0.4375	0.3163	0.5536	0.1354	0.1081
3	fixed per peak	0.5625	0.3171	0.5500	0.1643	0.1297
3	fixed per peak	0.6875	0.4489	0.5307	0.3492	0.2300
3	fixed per peak	0.8125	0.4150	0.5234	0.0679	0.2286
3	fixed per peak	0.9375	0.1813	0.5842	0.2886	0.3345
5	fixed per peak	0.0625	0.2917	0.7159	0.2885	0.3260
5	fixed per peak	0.1875	0.1247	0.5690	0.0677	0.1002
5	fixed per peak	0.3125	0.1599	0.3987	0.1638	0.0992
5	fixed per peak	0.4375	0.0925	0.2424	0.1354	0.0921
5	fixed per peak	0.5625	0.0927	0.2797	0.1643	0.1236
5	fixed per peak	0.6875	0.1601	0.4561	0.1747	0.1586
5	fixed per peak	0.8125	0.1248	0.6116	0.0679	0.0961
5	fixed per peak	0.9375	0.2916	0.7527	0.2886	0.3191
9	fixed per peak	0.0625	0.2947	0.7820	0.1439	0.2146
9	fixed per peak	0.1875	0.1120	0.6182	0.1495	0.0472
9	fixed per peak	0.3125	0.1475	0.4024	0.1621	0.0684
9	fixed per peak	0.4375	0.0835	0.1596	0.1354	0.0621
9	fixed per peak	0.5625	0.0836	0.2007	0.1507	0.0725
9	fixed per peak	0.6875	0.1478	0.4499	0.1538	0.1058
9	fixed per peak	0.8125	0.1123	0.6518	0.1417	0.1452
9	fixed per peak	0.9375	0.2946	0.8084	0.1774	0.1890
2	fixed total	0.0625	0.1682	0.5652	0.2885	0.3119
2	fixed total	0.1875	0.4616	0.6572	0.0677	0.1739
2	fixed total	0.3125	0.5166	0.7354	0.3494	0.4125

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Table 8: Condition-count paired profile-TVD effects with 95% intervals. Positive control, lookup, and mixture effects favor conditioning; a positive bank gap means the independent circuits have lower Training profile TVD.

K	Regime	no condition − conditioned		bank − conditioned		lookup − conditioned		mixture − conditioned	
2	fixed per peak	+0.0465	[+0.0289, +0.0682]	+0.0250	[+0.0221, +0.0274]	-0.0557	[-0.0697, -0.0417]	-0.0169	[-0.0309, -0.0029]
3	fixed per peak	+0.1754	[+0.1712, +0.1797]	+0.1140	[+0.1083, +0.1194]	-0.1263	[-0.1386, -0.1117]	-0.1155	[-0.1278, -0.1008]
5	fixed per peak	+0.2668	[+0.2587, +0.2737]	+0.0878	[+0.0860, +0.0897]	+0.0016	[-0.0012, +0.0043]	-0.0029	[-0.0053, -0.0006]
9	fixed per peak	+0.3354	[+0.3331, +0.3379]	+0.0464	[+0.0428, +0.0497]	-0.0077	[-0.0113, -0.0028]	-0.0464	[-0.0502, -0.0412]
2	fixed total	+0.0524	[+0.0339, +0.0724]	+0.0251	[+0.0226, +0.0273]	-0.0557	[-0.0693, -0.0424]	-0.0170	[-0.0305, -0.0036]
3	fixed total	+0.1756	[+0.1710, +0.1799]	+0.1140	[+0.1087, +0.1192]	-0.1261	[-0.1378, -0.1121]	-0.1152	[-0.1270, -0.1013]
5	fixed total	+0.2693	[+0.2620, +0.2739]	+0.0859	[+0.0823, +0.0892]	+0.0043	[+0.0001, +0.0091]	-0.0001	[-0.0040, +0.0042]
9	fixed total	+0.3333	[+0.3307, +0.3358]	+0.0482	[+0.0455, +0.0510]	-0.0104	[-0.0132, -0.0076]	-0.0491	[-0.0521, -0.0463]

K	Regime	c	Conditioned	No condition	Lookup	Mixture
2	fixed total	0.4375	0.3815	0.7669	0.5992	0.5613
2	fixed total	0.5625	0.3814	0.7618	0.5991	0.5613
2	fixed total	0.6875	0.5163	0.7272	0.3492	0.4125
2	fixed total	0.8125	0.4614	0.6647	0.0679	0.1741
2	fixed total	0.9375	0.1682	0.5744	0.2886	0.3121
3	fixed total	0.0625	0.1819	0.5630	0.2885	0.3341
3	fixed total	0.1875	0.4130	0.5417	0.0677	0.2213
3	fixed total	0.3125	0.4462	0.5575	0.3494	0.2114
3	fixed total	0.4375	0.3159	0.5515	0.1354	0.1081
3	fixed total	0.5625	0.3173	0.5480	0.1643	0.1297
3	fixed total	0.6875	0.4488	0.5325	0.3492	0.2300
3	fixed total	0.8125	0.4152	0.5262	0.0679	0.2286
3	fixed total	0.9375	0.1811	0.5862	0.2886	0.3345
5	fixed total	0.0625	0.2932	0.7198	0.2885	0.3260
5	fixed total	0.1875	0.1214	0.5725	0.0677	0.1002
5	fixed total	0.3125	0.1552	0.3974	0.1638	0.0992
5	fixed total	0.4375	0.0876	0.2355	0.1354	0.0921

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K	Regime	c	Conditioned	No condition	Lookup	Mixture
5	fixed total	0.5625	0.0876	0.2752	0.1643	0.1236
5	fixed total	0.6875	0.1562	0.4579	0.1747	0.1586
5	fixed total	0.8125	0.1222	0.6127	0.0679	0.0961
5	fixed total	0.9375	0.2926	0.7549	0.2886	0.3191
9	fixed total	0.0625	0.2936	0.7815	0.1439	0.2146
9	fixed total	0.1875	0.1150	0.6173	0.1495	0.0472
9	fixed total	0.3125	0.1519	0.4010	0.1621	0.0684
9	fixed total	0.4375	0.0872	0.1583	0.1354	0.0621
9	fixed total	0.5625	0.0878	0.2006	0.1507	0.0725
9	fixed total	0.6875	0.1532	0.4498	0.1538	0.1058
9	fixed total	0.8125	0.1157	0.6518	0.1417	0.1452
9	fixed total	0.9375	0.2933	0.8087	0.1774	0.1890

F Exhaustive paired intervals

The following machine-generated table lists every diagnostic comparison, not only the profile-TVD effects selected for the main text. The replication unit is always the matched optimizer seed; conditions are averaged before resampling.

Table 10: Paired intervals for $n = 8$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	-0.0243 [-0.0445, -0.0054]	unsupported
heldout lookup minus conditioned conditional mmd	-0.0005 [-0.0006, -0.0004]	unsupported
heldout lookup minus conditioned mean absolute error	+0.1103 [+0.0770, +0.1392]	supported
heldout lookup minus conditioned profile tvd	-0.0292 [-0.0468, -0.0108]	unsupported
heldout lookup minus conditioned width absolute error	-0.0962 [-0.1592, -0.0276]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0328 [-0.0540, -0.0126]	unsupported
heldout mixture minus conditioned conditional mmd	-0.0006 [-0.0008, -0.0005]	unsupported
heldout mixture minus conditioned mean absolute error	-0.0030 [-0.0251, +0.0171]	mixed
heldout mixture minus conditioned profile tvd	-0.0248 [-0.0418, -0.0064]	unsupported
heldout mixture minus conditioned width absolute error	-0.0194 [-0.0820, +0.0489]	mixed
heldout no condition minus conditioned bitstring tvd	+0.2735 [+0.2572, +0.2906]	supported
heldout no condition minus conditioned conditional mmd	+0.0406 [+0.0405, +0.0407]	supported
heldout no condition minus conditioned mean absolute error	+1.1205 [+1.0980, +1.1410]	supported
heldout no condition minus conditioned profile tvd	+0.2840 [+0.2695, +0.2990]	supported
heldout no condition minus conditioned width absolute error	+0.7802 [+0.6850, +0.8592]	supported
training conditioned minus single peak bitstring tvd	+0.0662 [+0.0474, +0.0856]	supported
training conditioned minus single peak conditional mmd	+0.0074 [+0.0073, +0.0076]	supported
training conditioned minus single peak mean absolute error	+0.1592 [+0.1460, +0.1735]	supported
training conditioned minus single peak profile tvd	+0.0923 [+0.0766, +0.1065]	supported
training conditioned minus single peak width absolute error	+0.0722 [+0.0179, +0.1220]	supported
training no condition minus conditioned bitstring tvd	+0.2388 [+0.2232, +0.2554]	supported
training no condition minus conditioned conditional mmd	+0.0525 [+0.0520, +0.0530]	supported
training no condition minus conditioned mean absolute error	+1.1017 [+1.0862, +1.1164]	supported
training no condition minus conditioned profile tvd	+0.2402 [+0.2278, +0.2536]	supported
training no condition minus conditioned width absolute error	+0.7978 [+0.7088, +0.8691]	supported

Table 11: Paired intervals for $n = 8$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	-0.0380 [-0.0491, -0.0277]	unsupported
heldout lookup minus conditioned conditional mmd	-0.0005 [-0.0006, -0.0003]	unsupported
heldout lookup minus conditioned mean absolute error	+0.0431 [+0.0287, +0.0570]	supported
heldout lookup minus conditioned profile tvd	-0.0280 [-0.0336, -0.0225]	unsupported
heldout lookup minus conditioned width absolute error	-0.0796 [-0.0909, -0.0692]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0362 [-0.0473, -0.0263]	unsupported
heldout mixture minus conditioned conditional mmd	-0.0003 [-0.0005, -0.0002]	unsupported
heldout mixture minus conditioned mean absolute error	-0.0377 [-0.0500, -0.0250]	unsupported
heldout mixture minus conditioned profile tvd	-0.0363 [-0.0432, -0.0292]	unsupported
heldout mixture minus conditioned width absolute error	-0.0028 [-0.0168, +0.0109]	mixed
heldout no condition minus conditioned bitstring tvd	+0.2375 [+0.2166, +0.2641]	supported
heldout no condition minus conditioned conditional mmd	+0.0407 [+0.0405, +0.0409]	supported
heldout no condition minus conditioned mean absolute error	+1.0973 [+1.0828, +1.1116]	supported
heldout no condition minus conditioned profile tvd	+0.2698 [+0.2542, +0.2965]	supported
heldout no condition minus conditioned width absolute error	+0.8385 [+0.7480, +0.8905]	supported
training conditioned minus single peak bitstring tvd	+0.0411 [+0.0310, +0.0525]	supported
training conditioned minus single peak conditional mmd	+0.0070 [+0.0069, +0.0071]	supported
training conditioned minus single peak mean absolute error	+0.1843 [+0.1766, +0.1915]	supported
training conditioned minus single peak profile tvd	+0.0788 [+0.0718, +0.0857]	supported
training conditioned minus single peak width absolute error	+0.0366 [+0.0221, +0.0507]	supported
training no condition minus conditioned bitstring tvd	+0.2055 [+0.1855, +0.2306]	supported
training no condition minus conditioned conditional mmd	+0.0514 [+0.0509, +0.0522]	supported
training no condition minus conditioned mean absolute error	+1.0859 [+1.0750, +1.0977]	supported
training no condition minus conditioned profile tvd	+0.2170 [+0.2015, +0.2418]	supported
training no condition minus conditioned width absolute error	+0.8941 [+0.7986, +0.9503]	supported

Table 12: Paired intervals for $n = 8$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	-0.0099 [-0.0165, -0.0032]	unsupported
heldout lookup minus conditioned conditional mmd	+0.0015 [+0.0014, +0.0016]	supported
heldout lookup minus conditioned mean absolute error	+0.1133 [+0.1009, +0.1227]	supported
heldout lookup minus conditioned profile tvd	-0.0276 [-0.0339, -0.0214]	unsupported
heldout lookup minus conditioned width absolute error	-0.0902 [-0.1037, -0.0750]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0596 [-0.0653, -0.0544]	unsupported
heldout mixture minus conditioned conditional mmd	-0.0005 [-0.0006, -0.0005]	unsupported
heldout mixture minus conditioned mean absolute error	-0.1080 [-0.1201, -0.0993]	unsupported
heldout mixture minus conditioned profile tvd	-0.0525 [-0.0584, -0.0461]	unsupported
heldout mixture minus conditioned width absolute error	-0.1657 [-0.1743, -0.1574]	unsupported
heldout no condition minus conditioned bitstring tvd	+0.1557 [+0.1278, +0.1969]	supported
heldout no condition minus conditioned conditional mmd	+0.0398 [+0.0395, +0.0404]	supported
heldout no condition minus conditioned mean absolute error	+1.0446 [+1.0327, +1.0538]	supported
heldout no condition minus conditioned profile tvd	+0.1900 [+0.1717, +0.2167]	supported
heldout no condition minus conditioned width absolute error	+0.5997 [+0.5068, +0.6518]	supported
training conditioned minus single peak bitstring tvd	+0.0503 [+0.0447, +0.0560]	supported
training conditioned minus single peak conditional mmd	+0.0067 [+0.0066, +0.0068]	supported
training conditioned minus single peak mean absolute error	+0.2277 [+0.2216, +0.2352]	supported
training conditioned minus single peak profile tvd	+0.0688 [+0.0635, +0.0742]	supported
training conditioned minus single peak width absolute error	+0.0995 [+0.0901, +0.1085]	supported
training no condition minus conditioned bitstring tvd	+0.1288 [+0.1018, +0.1699]	supported
training no condition minus conditioned conditional mmd	+0.0495 [+0.0490, +0.0506]	supported
training no condition minus conditioned mean absolute error	+1.0511 [+1.0412, +1.0592]	supported
training no condition minus conditioned profile tvd	+0.1460 [+0.1293, +0.1714]	supported
training no condition minus conditioned width absolute error	+0.7865 [+0.6961, +0.8358]	supported

Table 13: Paired intervals for $n = 12$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	-0.0371 [-0.0466, -0.0271]	unsupported
heldout lookup minus conditioned conditional mmd	-0.0004 [-0.0005, -0.0003]	unsupported
heldout lookup minus conditioned mean absolute error	+0.0992 [+0.0886, +0.1100]	supported
heldout lookup minus conditioned profile tvd	-0.0221 [-0.0255, -0.0190]	unsupported
heldout lookup minus conditioned width absolute error	-0.1215 [-0.1299, -0.1129]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0153 [-0.0247, -0.0051]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0000 [-0.0001, +0.0001]	mixed
heldout mixture minus conditioned mean absolute error	+0.0297 [+0.0162, +0.0431]	supported
heldout mixture minus conditioned profile tvd	-0.0016 [-0.0052, +0.0019]	mixed
heldout mixture minus conditioned width absolute error	+0.0261 [+0.0165, +0.0359]	supported
heldout no condition minus conditioned bitstring tvd	+0.3019 [+0.2894, +0.3164]	supported
heldout no condition minus conditioned conditional mmd	+0.0410 [+0.0409, +0.0411]	supported
heldout no condition minus conditioned mean absolute error	+1.7016 [+1.6905, +1.7132]	supported
heldout no condition minus conditioned profile tvd	+0.2892 [+0.2798, +0.2998]	supported
heldout no condition minus conditioned width absolute error	+0.5399 [+0.3070, +0.7984]	supported
training conditioned minus single peak bitstring tvd	+0.0509 [+0.0423, +0.0587]	supported
training conditioned minus single peak conditional mmd	+0.0074 [+0.0073, +0.0075]	supported
training conditioned minus single peak mean absolute error	+0.2118 [+0.2023, +0.2218]	supported
training conditioned minus single peak profile tvd	+0.0831 [+0.0791, +0.0872]	supported
training conditioned minus single peak width absolute error	+0.0752 [+0.0675, +0.0830]	supported
training no condition minus conditioned bitstring tvd	+0.2578 [+0.2494, +0.2684]	supported
training no condition minus conditioned conditional mmd	+0.0552 [+0.0541, +0.0562]	supported
training no condition minus conditioned mean absolute error	+1.6805 [+1.6708, +1.6909]	supported
training no condition minus conditioned profile tvd	+0.2419 [+0.2352, +0.2505]	supported
training no condition minus conditioned width absolute error	+0.5786 [+0.3458, +0.8372]	supported

Table 14: Paired intervals for $n = 12$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	-0.0356 [-0.0470, -0.0249]	unsupported
heldout lookup minus conditioned conditional mmd	-0.0001 [-0.0002, -0.0000]	unsupported
heldout lookup minus conditioned mean absolute error	+0.0568 [+0.0496, +0.0653]	supported
heldout lookup minus conditioned profile tvd	-0.0180 [-0.0213, -0.0146]	unsupported
heldout lookup minus conditioned width absolute error	-0.1159 [-0.1234, -0.1074]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0215 [-0.0328, -0.0107]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0001 [+0.0000, +0.0002]	supported
heldout mixture minus conditioned mean absolute error	-0.0563 [-0.0635, -0.0469]	unsupported
heldout mixture minus conditioned profile tvd	-0.0251 [-0.0283, -0.0216]	unsupported
heldout mixture minus conditioned width absolute error	+0.0116 [+0.0043, +0.0190]	supported
heldout no condition minus conditioned bitstring tvd	+0.3024 [+0.2842, +0.3197]	supported
heldout no condition minus conditioned conditional mmd	+0.0419 [+0.0415, +0.0424]	supported
heldout no condition minus conditioned mean absolute error	+1.6422 [+1.6358, +1.6508]	supported
heldout no condition minus conditioned profile tvd	+0.2793 [+0.2663, +0.2923]	supported
heldout no condition minus conditioned width absolute error	+0.8925 [+0.5774, +1.1911]	supported
training conditioned minus single peak bitstring tvd	+0.0312 [+0.0220, +0.0405]	supported
training conditioned minus single peak conditional mmd	+0.0071 [+0.0070, +0.0072]	supported
training conditioned minus single peak mean absolute error	+0.2773 [+0.2705, +0.2830]	supported
training conditioned minus single peak profile tvd	+0.0877 [+0.0839, +0.0911]	supported
training conditioned minus single peak width absolute error	+0.0343 [+0.0266, +0.0403]	supported
training no condition minus conditioned bitstring tvd	+0.2594 [+0.2388, +0.2782]	supported
training no condition minus conditioned conditional mmd	+0.0542 [+0.0524, +0.0560]	supported
training no condition minus conditioned mean absolute error	+1.6407 [+1.6322, +1.6518]	supported
training no condition minus conditioned profile tvd	+0.2291 [+0.2213, +0.2391]	supported
training no condition minus conditioned width absolute error	+1.0030 [+0.6903, +1.2993]	supported

Table 15: Paired intervals for $n = 12$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	+0.0156 [+0.0083, +0.0233]	supported
heldout lookup minus conditioned conditional mmd	+0.0018 [+0.0014, +0.0023]	supported
heldout lookup minus conditioned mean absolute error	+0.1243 [+0.0983, +0.1512]	supported
heldout lookup minus conditioned profile tvd	-0.0129 [-0.0151, -0.0107]	unsupported
heldout lookup minus conditioned width absolute error	-0.0644 [-0.0878, -0.0402]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0476 [-0.0501, -0.0453]	unsupported
heldout mixture minus conditioned conditional mmd	-0.0001 [-0.0001, -0.0000]	unsupported
heldout mixture minus conditioned mean absolute error	-0.1791 [-0.1851, -0.1731]	unsupported
heldout mixture minus conditioned profile tvd	-0.0548 [-0.0567, -0.0526]	unsupported
heldout mixture minus conditioned width absolute error	-0.2365 [-0.2563, -0.2149]	unsupported
heldout no condition minus conditioned bitstring tvd	+0.1953 [+0.1689, +0.2248]	supported
heldout no condition minus conditioned conditional mmd	+0.0408 [+0.0398, +0.0420]	supported
heldout no condition minus conditioned mean absolute error	+1.5417 [+1.5374, +1.5463]	supported
heldout no condition minus conditioned profile tvd	+0.2019 [+0.1903, +0.2138]	supported
heldout no condition minus conditioned width absolute error	+0.8031 [+0.5245, +1.0192]	supported
training conditioned minus single peak bitstring tvd	+0.0288 [+0.0256, +0.0321]	supported
training conditioned minus single peak conditional mmd	+0.0066 [+0.0066, +0.0066]	supported
training conditioned minus single peak mean absolute error	+0.3499 [+0.3473, +0.3527]	supported
training conditioned minus single peak profile tvd	+0.0582 [+0.0554, +0.0613]	supported
training conditioned minus single peak width absolute error	+0.0797 [+0.0647, +0.0938]	supported
training no condition minus conditioned bitstring tvd	+0.1687 [+0.1409, +0.1999]	supported
training no condition minus conditioned conditional mmd	+0.0500 [+0.0480, +0.0524]	supported
training no condition minus conditioned mean absolute error	+1.5664 [+1.5605, +1.5728]	supported
training no condition minus conditioned profile tvd	+0.1608 [+0.1515, +0.1722]	supported
training no condition minus conditioned width absolute error	+1.0492 [+0.7734, +1.2647]	supported

Table 16: Paired intervals for $n = 16$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	-0.0429 [-0.0538, -0.0319]	unsupported
heldout lookup minus conditioned conditional mmd	-0.0001 [-0.0001, +0.0000]	mixed
heldout lookup minus conditioned mean absolute error	+0.1446 [+0.1381, +0.1499]	supported
heldout lookup minus conditioned profile tvd	-0.0229 [-0.0279, -0.0174]	unsupported
heldout lookup minus conditioned width absolute error	-0.1356 [-0.1557, -0.1146]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0060 [-0.0166, +0.0048]	mixed
heldout mixture minus conditioned conditional mmd	+0.0004 [+0.0003, +0.0005]	supported
heldout mixture minus conditioned mean absolute error	+0.0464 [+0.0369, +0.0546]	supported
heldout mixture minus conditioned profile tvd	+0.0117 [+0.0067, +0.0172]	supported
heldout mixture minus conditioned width absolute error	+0.0987 [+0.0783, +0.1208]	supported
heldout no condition minus conditioned bitstring tvd	+0.3087 [+0.2927, +0.3256]	supported
heldout no condition minus conditioned conditional mmd	+0.0399 [+0.0397, +0.0400]	supported
heldout no condition minus conditioned mean absolute error	+2.2786 [+2.2711, +2.2848]	supported
heldout no condition minus conditioned profile tvd	+0.3139 [+0.3059, +0.3231]	supported
heldout no condition minus conditioned width absolute error	+0.6858 [+0.2739, +1.1408]	supported
training conditioned minus single peak bitstring tvd	+0.0464 [+0.0370, +0.0554]	supported
training conditioned minus single peak conditional mmd	+0.0068 [+0.0067, +0.0069]	supported
training conditioned minus single peak mean absolute error	+0.2799 [+0.2737, +0.2868]	supported
training conditioned minus single peak profile tvd	+0.0835 [+0.0785, +0.0878]	supported
training conditioned minus single peak width absolute error	+0.0675 [+0.0499, +0.0838]	supported
training no condition minus conditioned bitstring tvd	+0.2515 [+0.2406, +0.2632]	supported
training no condition minus conditioned conditional mmd	+0.0538 [+0.0525, +0.0550]	supported
training no condition minus conditioned mean absolute error	+2.2557 [+2.2469, +2.2646]	supported
training no condition minus conditioned profile tvd	+0.2472 [+0.2421, +0.2519]	supported
training no condition minus conditioned width absolute error	+0.7318 [+0.3236, +1.1828]	supported

Table 17: Paired intervals for $n = 16$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	-0.0418 [-0.0510, -0.0318]	unsupported
heldout lookup minus conditioned conditional mmd	+0.0001 [+0.0001, +0.0002]	supported
heldout lookup minus conditioned mean absolute error	+0.0876 [+0.0800, +0.0953]	supported
heldout lookup minus conditioned profile tvd	-0.0040 [-0.0071, +0.0001]	mixed
heldout lookup minus conditioned width absolute error	-0.1386 [-0.1487, -0.1256]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0193 [-0.0282, -0.0093]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0004 [+0.0003, +0.0005]	supported
heldout mixture minus conditioned mean absolute error	-0.0697 [-0.0805, -0.0566]	unsupported
heldout mixture minus conditioned profile tvd	-0.0099 [-0.0133, -0.0057]	unsupported
heldout mixture minus conditioned width absolute error	+0.0298 [+0.0163, +0.0472]	supported
heldout no condition minus conditioned bitstring tvd	+0.3069 [+0.2923, +0.3265]	supported
heldout no condition minus conditioned conditional mmd	+0.0408 [+0.0403, +0.0413]	supported
heldout no condition minus conditioned mean absolute error	+2.1984 [+2.1900, +2.2077]	supported
heldout no condition minus conditioned profile tvd	+0.3138 [+0.3019, +0.3270]	supported
heldout no condition minus conditioned width absolute error	+0.9599 [+0.4548, +1.4967]	supported
training conditioned minus single peak bitstring tvd	+0.0352 [+0.0264, +0.0432]	supported
training conditioned minus single peak conditional mmd	+0.0067 [+0.0066, +0.0068]	supported
training conditioned minus single peak mean absolute error	+0.3687 [+0.3605, +0.3757]	supported
training conditioned minus single peak profile tvd	+0.0933 [+0.0895, +0.0964]	supported
training conditioned minus single peak width absolute error	+0.0278 [+0.0091, +0.0434]	supported
training no condition minus conditioned bitstring tvd	+0.2575 [+0.2402, +0.2765]	supported
training no condition minus conditioned conditional mmd	+0.0534 [+0.0513, +0.0554]	supported
training no condition minus conditioned mean absolute error	+2.2020 [+2.1924, +2.2116]	supported
training no condition minus conditioned profile tvd	+0.2440 [+0.2354, +0.2537]	supported
training no condition minus conditioned width absolute error	+1.1301 [+0.6214, +1.6714]	supported

Table 18: Paired intervals for $n = 16$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	+0.0250 [+0.0190, +0.0313]	supported
heldout lookup minus conditioned conditional mmd	+0.0029 [+0.0026, +0.0032]	supported
heldout lookup minus conditioned mean absolute error	+0.2461 [+0.2180, +0.2729]	supported
heldout lookup minus conditioned profile tvd	-0.0192 [-0.0205, -0.0178]	unsupported
heldout lookup minus conditioned width absolute error	-0.1705 [-0.1893, -0.1523]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0559 [-0.0594, -0.0518]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0001 [+0.0001, +0.0002]	supported
heldout mixture minus conditioned mean absolute error	-0.2429 [-0.2518, -0.2340]	unsupported
heldout mixture minus conditioned profile tvd	-0.0681 [-0.0696, -0.0666]	unsupported
heldout mixture minus conditioned width absolute error	-0.3812 [-0.4014, -0.3606]	unsupported
heldout no condition minus conditioned bitstring tvd	+0.2406 [+0.2034, +0.2781]	supported
heldout no condition minus conditioned conditional mmd	+0.0405 [+0.0394, +0.0417]	supported
heldout no condition minus conditioned mean absolute error	+2.0498 [+2.0416, +2.0579]	supported
heldout no condition minus conditioned profile tvd	+0.1981 [+0.1780, +0.2176]	supported
heldout no condition minus conditioned width absolute error	+0.8736 [+0.4476, +1.2787]	supported
training conditioned minus single peak bitstring tvd	+0.0258 [+0.0219, +0.0293]	supported
training conditioned minus single peak conditional mmd	+0.0064 [+0.0064, +0.0064]	supported
training conditioned minus single peak mean absolute error	+0.4633 [+0.4530, +0.4731]	supported
training conditioned minus single peak profile tvd	+0.0523 [+0.0495, +0.0553]	supported
training conditioned minus single peak width absolute error	+0.1339 [+0.1182, +0.1497]	supported
training no condition minus conditioned bitstring tvd	+0.2100 [+0.1710, +0.2486]	supported
training no condition minus conditioned conditional mmd	+0.0499 [+0.0473, +0.0526]	supported
training no condition minus conditioned mean absolute error	+2.1053 [+2.0940, +2.1171]	supported
training no condition minus conditioned profile tvd	+0.1667 [+0.1570, +0.1770]	supported
training no condition minus conditioned width absolute error	+1.1659 [+0.7386, +1.5723]	supported

Table 19: Paired intervals for $n = 19$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	-0.0610 [-0.0776, -0.0458]	unsupported
heldout lookup minus conditioned conditional mmd	-0.0001 [-0.0003, +0.0001]	mixed
heldout lookup minus conditioned mean absolute error	+0.1758 [+0.1493, +0.2119]	supported
heldout lookup minus conditioned profile tvd	-0.0217 [-0.0240, -0.0192]	unsupported
heldout lookup minus conditioned width absolute error	-0.1371 [-0.1597, -0.1162]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0175 [-0.0342, -0.0021]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0003 [+0.0001, +0.0005]	supported
heldout mixture minus conditioned mean absolute error	+0.0533 [+0.0228, +0.0942]	supported
heldout mixture minus conditioned profile tvd	+0.0188 [+0.0164, +0.0213]	supported
heldout mixture minus conditioned width absolute error	+0.1641 [+0.1409, +0.1852]	supported
heldout no condition minus conditioned bitstring tvd	+0.2943 [+0.2769, +0.3101]	supported
heldout no condition minus conditioned conditional mmd	+0.0389 [+0.0386, +0.0391]	supported
heldout no condition minus conditioned mean absolute error	+2.7073 [+2.6799, +2.7452]	supported
heldout no condition minus conditioned profile tvd	+0.3356 [+0.3282, +0.3468]	supported
heldout no condition minus conditioned width absolute error	+0.4266 [+0.1216, +0.9637]	supported
training conditioned minus single peak bitstring tvd	+0.0620 [+0.0488, +0.0763]	supported
training conditioned minus single peak conditional mmd	+0.0065 [+0.0064, +0.0067]	supported
training conditioned minus single peak mean absolute error	+0.3355 [+0.3104, +0.3554]	supported
training conditioned minus single peak profile tvd	+0.0870 [+0.0847, +0.0893]	supported
training conditioned minus single peak width absolute error	+0.0574 [+0.0395, +0.0769]	supported
training no condition minus conditioned bitstring tvd	+0.2394 [+0.2256, +0.2523]	supported
training no condition minus conditioned conditional mmd	+0.0533 [+0.0519, +0.0542]	supported
training no condition minus conditioned mean absolute error	+2.6746 [+2.6561, +2.6983]	supported
training no condition minus conditioned profile tvd	+0.2620 [+0.2594, +0.2645]	supported
training no condition minus conditioned width absolute error	+0.4747 [+0.1704, +1.0099]	supported

Table 20: Paired intervals for $n = 19$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	-0.0606 [-0.0749, -0.0461]	unsupported
heldout lookup minus conditioned conditional mmd	+0.0002 [+0.0002, +0.0003]	supported
heldout lookup minus conditioned mean absolute error	+0.1145 [+0.1000, +0.1286]	supported
heldout lookup minus conditioned profile tvd	+0.0016 [-0.0012, +0.0043]	mixed
heldout lookup minus conditioned width absolute error	-0.1451 [-0.1555, -0.1358]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0300 [-0.0443, -0.0155]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0005 [+0.0005, +0.0006]	supported
heldout mixture minus conditioned mean absolute error	-0.0696 [-0.0804, -0.0585]	unsupported
heldout mixture minus conditioned profile tvd	-0.0029 [-0.0053, -0.0006]	unsupported
heldout mixture minus conditioned width absolute error	+0.0521 [+0.0416, +0.0622]	supported
heldout no condition minus conditioned bitstring tvd	+0.3037 [+0.2885, +0.3183]	supported
heldout no condition minus conditioned conditional mmd	+0.0400 [+0.0394, +0.0404]	supported
heldout no condition minus conditioned mean absolute error	+2.6224 [+2.6098, +2.6350]	supported
heldout no condition minus conditioned profile tvd	+0.3360 [+0.3223, +0.3507]	supported
heldout no condition minus conditioned width absolute error	+0.8220 [+0.2164, +1.5105]	supported
training conditioned minus single peak bitstring tvd	+0.0428 [+0.0296, +0.0560]	supported
training conditioned minus single peak conditional mmd	+0.0064 [+0.0063, +0.0064]	supported
training conditioned minus single peak mean absolute error	+0.4279 [+0.4206, +0.4354]	supported
training conditioned minus single peak profile tvd	+0.0878 [+0.0860, +0.0897]	supported
training conditioned minus single peak width absolute error	+0.0319 [+0.0135, +0.0514]	supported
training no condition minus conditioned bitstring tvd	+0.2559 [+0.2370, +0.2723]	supported
training no condition minus conditioned conditional mmd	+0.0530 [+0.0507, +0.0549]	supported
training no condition minus conditioned mean absolute error	+2.6314 [+2.6155, +2.6474]	supported
training no condition minus conditioned profile tvd	+0.2668 [+0.2587, +0.2737]	supported
training no condition minus conditioned width absolute error	+1.0212 [+0.4158, +1.7108]	supported

Table 21: Paired intervals for $n = 19$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout lookup minus conditioned bitstring tvd	+0.0074 [-0.0032, +0.0165]	mixed
heldout lookup minus conditioned conditional mmd	+0.0020 [+0.0016, +0.0023]	supported
heldout lookup minus conditioned mean absolute error	+0.2478 [+0.2051, +0.2986]	supported
heldout lookup minus conditioned profile tvd	-0.0111 [-0.0165, -0.0049]	unsupported
heldout lookup minus conditioned width absolute error	-0.1981 [-0.2308, -0.1643]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0656 [-0.0768, -0.0563]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0002 [-0.0002, +0.0004]	mixed
heldout mixture minus conditioned mean absolute error	-0.2483 [-0.2763, -0.2077]	unsupported
heldout mixture minus conditioned profile tvd	-0.0619 [-0.0670, -0.0560]	unsupported
heldout mixture minus conditioned width absolute error	-0.4633 [-0.4961, -0.4286]	unsupported
heldout no condition minus conditioned bitstring tvd	+0.2272 [+0.1839, +0.2702]	supported
heldout no condition minus conditioned conditional mmd	+0.0401 [+0.0384, +0.0418]	supported
heldout no condition minus conditioned mean absolute error	+2.4747 [+2.4455, +2.5170]	supported
heldout no condition minus conditioned profile tvd	+0.2261 [+0.2088, +0.2418]	supported
heldout no condition minus conditioned width absolute error	+0.8519 [+0.1776, +1.5129]	supported
training conditioned minus single peak bitstring tvd	+0.0385 [+0.0306, +0.0474]	supported
training conditioned minus single peak conditional mmd	+0.0063 [+0.0062, +0.0066]	supported
training conditioned minus single peak mean absolute error	+0.5383 [+0.5122, +0.5573]	supported
training conditioned minus single peak profile tvd	+0.0603 [+0.0544, +0.0659]	supported
training conditioned minus single peak width absolute error	+0.1626 [+0.1321, +0.1922]	supported
training no condition minus conditioned bitstring tvd	+0.1940 [+0.1476, +0.2405]	supported
training no condition minus conditioned conditional mmd	+0.0503 [+0.0465, +0.0542]	supported
training no condition minus conditioned mean absolute error	+2.5272 [+2.5060, +2.5506]	supported
training no condition minus conditioned profile tvd	+0.1807 [+0.1648, +0.1999]	supported
training no condition minus conditioned width absolute error	+1.1021 [+0.3924, +1.7942]	supported

G Archived Gaussian ablations

The topology, encoding, and ancilla studies predate the frozen factorial and are preserved as bounded ablations. Their metrics are regenerated from the archived JSON/NPZ pairs and are not pooled with the new campaign. Complete connectivity uses 190 phases at $n = 19$; the lattice and heavy-hex banks use 48 and 39. The symmetric-reupload comparison uses three matched seeds. The ancilla ladder uses $m \in \{0, 1, 2, 3, 5, 7\}$ and compares each augmented model with a visible-only matched-parameter control.

Permutation twirling replaces a raw sample x by $\pi(x)$ for a uniformly random qubit permutation π. For an exchangeable target, this preserves the Hamming-weight profile and makes the output uniform within each observed shell in expectation. It is therefore a legitimate target-symmetry-aware post-processing baseline. It is not a symmetry learned by the circuit, it uses known target structure, and it is excluded from raw-circuit claims.

H Negative two-coordinate feasibility study

An archived eight-qubit pilot attempted to condition jointly on Gaussian location and width. Its frozen design required single-condition representability before any shared-model or interpolation interpretation. At seed 0, the four single-condition bitstring TVDs were 0.3883, 0.3177, 0.2220, and 0.0945. The two narrow-width cells failed the declared capacity screen; their width errors were 0.4659 and 0.1058. The study stopped, so it supplies no evidence for two-coordinate conditional generation. We include it to mark the boundary of the present one-coordinate claim and to prevent selective reporting of only successful Gaussian variants.

I Campaign integrity and reproduction

The Gaussian campaign is isolated under `results/gaussian-foundations-v1`. The immutable manifest digest binds the run plan; a source-tree digest binds the numerical core used by the resource preflight and every result, while a companion dependency-closure snapshot binds the encoder, ansatz, simulator, target, artifact, environment, and lockfile sources. Dataset archives record role, derived seed, condition grid, master-pool hash, and split hash. A result is complete only when its envelope, checkpoint, and probability archive all exist and their digests agree. Reservations without a complete optimizer state are never silently restarted.

From the repository root, the principal commands are:

```
uv sync --frozen
uv run python scripts/prepare_gaussian_campaign.py
uv run python scripts/run_gaussian_campaign.py preflight
uv run python scripts/freeze_gaussian_execution_source.py
uv run python scripts/run_gaussian_campaign.py run \
  --shard-count 5 --shard-index 0
uv run python scripts/summarize_gaussian_campaign.py
uv run python scripts/verify_gaussian_legacy.py
uv run python scripts/finalize_gaussian_claims.py
uv run python scripts/generate_gaussian_tables.py
uv run python scripts/plot_gaussian_campaign.py
uv run pytest -q
uv run ruff check src scripts tests
uv run python scripts/build_gaussian_arxiv_archive.py
uv run python scripts/audit_gaussian_release.py
```

Shard indices 1–4 are executed analogously. Re-running a completed shard verifies and skips immutable outputs. Report, claim, table, and figure generators refuse partial campaign state.

The arXiv source builder copies only the Gaussian manuscript sources, generated tables, five figures, bibliography, and required style files into a temporary directory. It rejects forbidden topic strings, compiles twice, checks citations and references, renders every page, and records SHA-256 digests for the source archive and PDF. The upload-ready flag remains false until the team supplies and verifies all author and release metadata and performs a final human proof.